Chapter 6
Viewshed Analysis

Viewshed (visibility) analysis is used in many different fields for both practical and aesthetic applications. It can play an important role when planning new buildings or roads especially in urban settings where obstructed views may raise safety concerns. In recreation areas views of beautiful landscapes are highly valued and protected with great passion. Visibility analysis is also crucial when planning location of monitoring cameras or communication towers in order to maximize coverage. With the increasing availability of high-resolution DEMs and DSMs derived from lidar visibility analysis is becoming more accurate, broading the range of its applications. We used Tangible Landscape to analyze viewsheds on North Carolina State University's (NCSU) Centennial Campus from different observer positions and explored how future development would affect the viewsheds. We introduced object recognition to collaboratively designate observer positions.

6.1 Line of Sight Analysis

Line-of-sight analysis is used to map visible areas. If the line of sight between point A and point B does not intersect the terrain then these points are mutually visible. More precisely, with the slope of the line between A and B defined as $(z_B - z_A)/d_{AB}$ where z is the height and d is the horizontal distance, points A and B are considered visible to each other if the line does not cross any location C such that the slope of AC is larger than the slope of AB (Fig. 6.1). The viewshed of A on a raster DEM is the set of all cells of the DEM that are visible from A. The height of the observer A above terrain has significant influence on the viewshed extent. The observer height can be changed to model the view of a standing person, the view from a multi-story building, or the view from the top of a cell tower. The resulting viewshed can be represented for example as a binary raster (containing only 0 and 1s) or a raster with the values of vertical angles with regard to the viewpoint.

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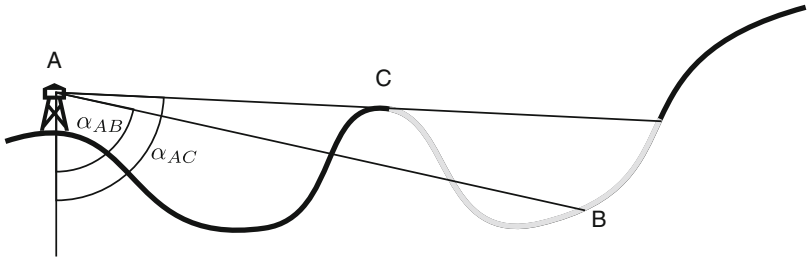

Fig. 6.1 Line of sight analysis: point B is not visible from point A because the line of sight between A and B has a smaller slope $\tan \alpha_{AB}$ than the line of sight between A and C

Viewsheds from multiple locations can be computed and combined as the sum of binary viewsheds into a cumulative viewshed (Wheatley 1995) that allows us to assess the overall visibility of a place from selected locations. The concepts of visibility analysis have been applied to diverse geospatial applications. For example landform identification using geomorphons combines visibility analysis with computer vision (Jasiewicz and Stepinski 2013) (see Sect. 4.1.3). Several terrain relief visualization techniques take advantage of visibility analysis to highlight subtle terrain features. For example sky-view factor—the visible part of the sky unobscured by relief—can be used to render topographic relief as if it was diffusely illuminated in order to clearly visualize the relative height of features. Sky-view factor uses openness as a proxy for uniform diffuse illumination (Zakšek et al. 2011). It is implemented in GRASS GIS as the add-on module *r.skyview*. Haverkort et al. (2009) describe an efficient method implemented in the module *r.viewshed* for computing viewsheds on a raster DEM using a technique called line sweeping (Fig. 6.1).

6.2 Case Study: Viewsheds Around Lake Raleigh

We used object recognition with Tangible Landscape to explore different viewsheds on NCSU's Centennial Campus by placing markers representing observer positions on a physical model of the landscape. We also evaluated how changes to the topography, the forest canopy, and the buildings would impact these viewsheds.

6.2.1 Site Description and Model

Our study area shown in Fig. 6.2 is the Lake Raleigh landscape on NCSU's Centennial Campus. Key features of the landscape include Lake Raleigh Woods in the southwest, the Chancellor's house and State Club in the south, and the Hunt Library in the northwest.

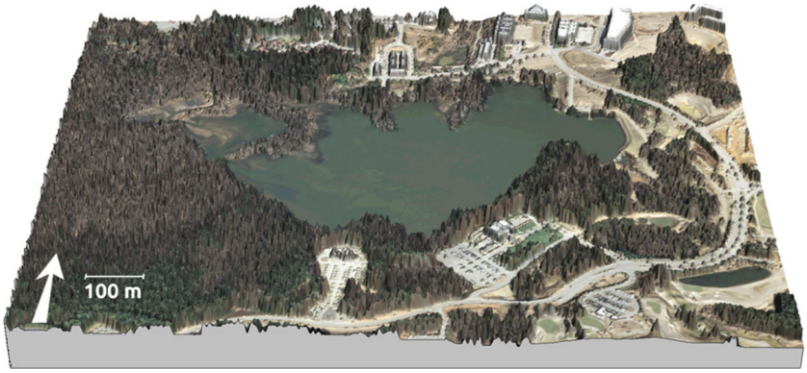

Fig. 6.2 Lake Raleigh study site: a lidar-based DSM with an orthophoto draped over the surface

We CNC routed a base model and a mold of the landscape surface from MDF based on a lidar-derived DEM and DSM. Both models are approximately 1:2400 scale without vertical exaggeration. We used the base and the mold to cast a malleable layer of polymer enriched sand representing the forest canopy and buildings on top of the terrain model.

6.2.2 Identifying View Points

We modeled the views from different locations by placing markers on the model for Tangible Landscape to detect and recognize. The position of each marker was registered as a vector point and its coordinates were then used as an input for automated computation of viewshed from this location.

Given the complex 3D form and steep slopes of some terrains it may be hard to place markers on a rigid model such as a 3D printed or CNC routed terrain. If the model is coated with magnetic paint then a magnetized marker will hold to the slopes, normal to the surface (see Fig. 3.7). If the model has been sculpted or cast in polymeric sand then markers can simply be stuck vertically into the sand.

There are multiple ways to detect markers. The most reliable approach is to detect the markers based on the difference between a base scan of the terrain model without the markers and the scan with the markers placed on the model. We computed the difference using the function defined in the code snippet in Sect. 4.2.2. A threshold value for the difference in elevation must be set in order to distinguish markers from noise. The threshold can be specified in millimeters and since we know the horizontal scale and vertical exaggeration during the processing it can be automatically converted into the real world elevation difference required in the following code snippet. Areas with difference values above the threshold are vectorized as points for viewshed analysis as follows. We first convert the areas from

raster to vector areas (polygons) using the module *r.to.vect* and then extract the area centroids as vector points using the module *v.to.points*.

```
adjust_scan('scan_before', 'scan_after', 'scan_adjusted')
gscript.mapcalc("change = if(scan_adjusted - scan_before >
    threshold, 1, null())")
gscript.run_command('r.to.vect', flags='st', input='change',
    output='change_vector', type='area')
gscript.run_command('v.to.points', flags='t',
    input='change_vector', type='centroid', output='markers')
```

6.2.3 Visibility Analysis on DSM Using Markers

We explored viewsheds from different points of interests such as the Chancellor's house, the North Shore neighborhood, the Hunt Library's observation deck, the lakeside pier, and the middle of the lake. The viewsheds were computed on the lidar-derived DSM. To identify the markers we first scanned the model and saved the scanned raster. We continued scanning while placing the markers at the locations of interest (Fig. 6.3). The markers were then identified and stored as vector points. We used the following workflow to compute the viewsheds and set the desired color scheme. In this case visible areas are highlighted in yellow.

```
coordinates = gscript.read_command('v.out.ascii',
    input='change_vector', separator=',').strip().splitlines()
for i, line in coordinates:
   output = 'viewshed_%s' % i
   coordinate = [float(c) for c in line.split(',')[0:2]]
   gscript.run_command('r.viewshed', flags='b',
       input='elevation', output=output,
       observer_elevation=1.75, coordinates=coordinate)
   gscript.run_command('r.null', map=output, setnull=0)
   gscript.write_command('r.colors', map=output, rules='-',
       stdin='1 yellow')
```

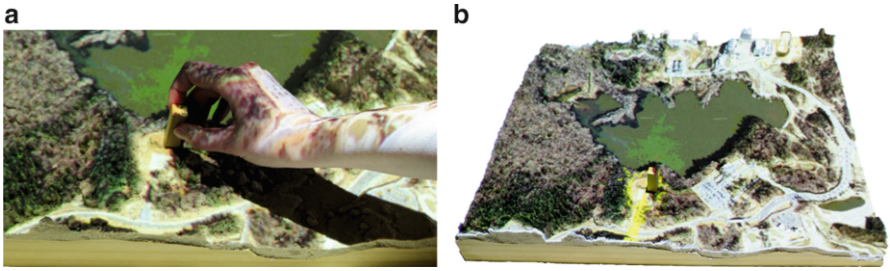

Fig. 6.3 Placing a marker to identify a viewpoint

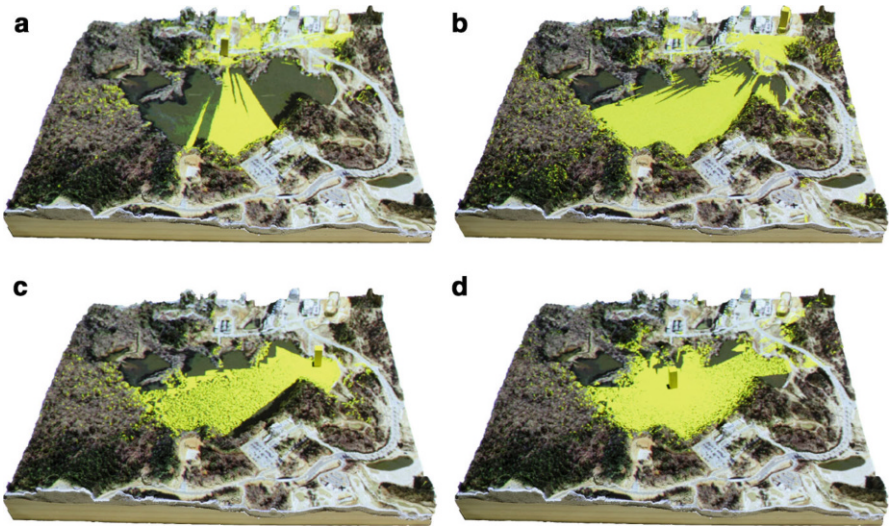

Fig. 6.4 Viewsheds computed on lidar-based DSM from viewpoints at (**a**) the North Shore neighborhood, (**b**) Hunt Library's observation deck, (**c**) a lakeside pier, (**d**) and a canoe on the lake

The viewsheds computed from different marker positions are shown in Fig. 6.4. The best views of the lake are from the Hunt Library's observation deck and the lakeside pier.

6.2.4 Modeling Viewsheds from a New Building

In the previous section we computed viewsheds on the lidar-derived DSM. In this section we used the scanned surface to model viewsheds on a modified landscape. We explored a scenario in which we built a hotel on the southeastern side of the lake and assessed the viewshed from the new hotel (Fig. 6.5a).

We began by sculpting the new building in sand. Then we scanned the modified sand surface and saved the scan. We placed a marker that was automatically detected and digitized as a viewpoint. We computed the viewshed using the base scan rather than the newer scan with the marker because the marker would act as an obstacle blocking the observer's view. Fig. 6.5c shows the viewshed from the southwest side of the new hotel. The hotel is oriented so that it has nice views of the lake. Since the hotel is a tall building it is visible from many places around the lake such as Hunt Library (Fig. 6.5d).

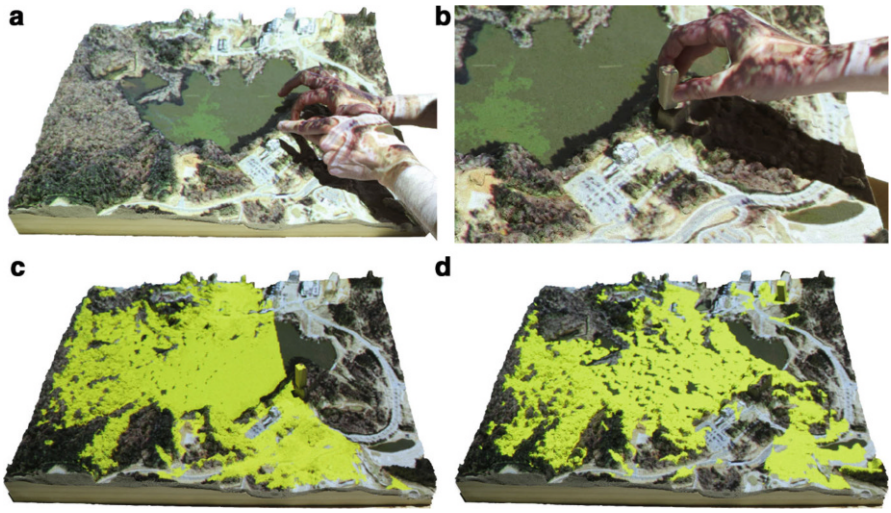

Fig. 6.5 Modeling viewsheds from a new building: (**a**) building a new hotel, (**b**) digitizing a viewpoint, and evaluating viewsheds from (**c**) the hotel and (**d**) the library

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